

UMPHANG, Thailand

WMO: 483850

Lat: 16.0253N

Lon: 98.8600E

Elev: 460

StdP: 95.92

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	11.6	7.1	6.6	20.0	8.9	7.5	19.0	3.7	28.2	3.2	28.2	0.0	N/A	0.226

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	13.5	35.6	21.7	34.5	21.6	33.4	21.8	25.2	29.8	24.8	29.3	24.6	29.0	0.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	20.0	26.9	23.7	19.6	26.5	23.5	19.3	26.3	79.9	30.1	78.4	29.5	77.4	29.3	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.1	2.1	0.9	DB	7.4	37.3	2.1	1.0	5.9	38.0	4.7	38.6	3.5	39.1	2.0	39.8	(4)
(5)			WB	6.9	27.1	2.1	1.3	5.4	28.0	4.2	28.8	3.1	29.5	1.6	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.9	21.8	23.5	25.6	27.3	27.0	26.2	25.3	25.2	25.6	25.4	23.8	21.9	(6)	
	DBStd	2.26	2.04	1.80	1.55	1.28	1.32	1.14	0.98	0.94	0.96	1.03	1.76	2.22	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	6	2	1	0	0	0	0	0	0	0	0	1	2	(9)	
	CDD10.0	5433	366	378	482	518	527	485	475	472	468	478	414	369	(10)	
	CDD18.3	2397	110	145	224	268	269	235	217	213	218	219	165	113	(11)	
	CDH23.3	18639	1077	1578	2354	2631	2229	1623	1210	1107	1268	1423	1211	928	(12)	
(13) Wind	CDH26.7	7021	393	754	1248	1368	898	454	233	194	304	462	426	287	(13)	
	WSAvg	0.1	0.1	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.0	0.1	0.1	0.1	(14)	
	PrecAvg	1494	9	10	53	90	192	195	251	254	258	147	27	9	(15)	
	PrecMax	1927	33	65	136	300	373	296	521	397	430	327	134	75	(16)	
	PrecMin	961	0	0	1	12	69	96	143	140	101	8	0	0	(17)	
	PrecStd	240	12	16	46	64	80	57	86	63	79	72	31	17	(18)	
															(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.4	34.7	36.2	37.2	36.1	33.0	31.2	30.9	31.8	32.2	32.2	31.8	(19)	
		MCWB	20.4	20.1	20.4	22.3	23.6	24.3	24.2	24.5	24.5	23.8	23.3	22.1	(20)	
	2%	DB	31.3	33.6	35.1	36.0	34.0	31.7	30.0	29.8	30.5	31.2	31.3	30.6	(21)	
		MCWB	20.5	20.2	20.4	22.3	23.9	24.3	24.0	24.3	24.4	23.8	22.9	21.7	(22)	
	5%	DB	30.3	32.7	34.2	34.7	32.5	30.4	28.9	28.7	29.6	30.4	30.5	29.7	(23)	
		MCWB	20.3	20.4	20.4	22.6	24.0	24.2	23.8	24.0	24.2	23.6	22.6	21.4	(24)	
	10%	DB	29.2	31.5	33.1	33.3	31.2	29.2	28.0	27.7	28.5	29.5	29.5	28.4	(25)	
		MCWB	19.9	20.2	20.5	22.9	24.1	24.0	23.7	23.7	24.0	23.5	22.3	20.8	(26)	
	0.4%	WB	23.3	23.6	23.8	25.2	25.5	25.4	25.0	25.2	25.3	25.2	24.8	24.1	(27)	
		MCDB	28.8	30.6	30.5	31.6	30.7	29.8	28.7	28.9	29.5	29.9	29.9	29.1	(28)	
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	22.3	22.6	23.1	24.5	25.0	24.9	24.6	24.6	24.8	24.7	24.2	23.1	(29)	
		MCDB	27.9	29.7	30.2	31.0	30.5	29.1	28.1	28.1	28.8	29.2	29.2	28.3	(30)	
	5%	WB	21.7	21.9	22.6	24.1	24.6	24.6	24.2	24.2	24.5	24.3	23.6	22.3	(31)	
		MCDB	27.2	28.9	29.4	30.7	29.8	28.5	27.5	27.5	28.3	28.6	28.5	27.4	(32)	
	10%	WB	21.0	21.3	22.1	23.6	24.3	24.2	23.9	24.0	24.2	23.9	23.0	21.6	(33)	
		MCDB	26.5	28.2	28.4	30.1	29.2	28.0	27.1	27.0	27.7	28.0	27.5	26.6	(34)	
	5% DB	MDBR	16.2	17.5	16.2	13.5	9.2	6.9	5.9	5.9	6.8	9.1	12.0	14.2	(35)	
		MCDBR	17.3	18.7	18.1	15.6	11.4	8.7	7.4	7.3	8.2	10.5	13.0	14.6	(36)	
		MCWBR	7.3	6.7	5.2	4.1	3.2	2.6	2.5	2.7	3.0	4.1	5.6	6.4	(37)	
		MCWBR	13.7	15.5	14.0	12.7	9.4	7.5	6.4	6.5	7.4	8.9	11.3	12.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.448	0.534	0.657	0.585	0.469	0.422	0.416	0.419	0.437	0.510	0.450	0.443	(40)	
		taud	2.160	1.945	1.675	1.902	2.279	2.434	2.449	2.429	2.362	2.103	2.245	2.201	(41)	
	Ebn at Noon		837	780	700	754	834	866	872	878	866	792	830	829	(42)	
		Edh at Noon	145	188	253	202	136	116	114	118	125	158	132	136	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.75	5.27	5.50	5.88	5.07	4.34	3.98	3.96	4.38	4.66	4.81	4.62	(44)	
	RadStd		0.21	0.22	0.35	0.35	0.47	0.43	0.41	0.33	0.30	0.27	0.27	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	+0.13	+0.20	N/A	N/A	N/A	N/A

Nomenclature: See separate page